Aryan Shrivastava

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4th-year Integrated M.Sc student, School of Physical Sciences,
National Institute of Science Education and Research (NISER)
Bhubaneswar, Odisha, India

Objective

To work in scientific environments where I can apply my skills in physics and machine learning. Passionate about experiments and enjoy solving complex problems. Aim to contribute to innovative research and gain valuable hands-on experience.

Awards & Achievements

- Recipient of the Innovation in Science Pursuit for Inspired Research (INSPIRE) scholarship.
- Silver medal, Undergraduate category, National Competition for Computational Physics (NCICP 2025), organized by the Indian Association of Physics Teachers (IAPT).
- Gold medal, Postgraduate category, NCICP 2026, organized by the IAPT.
- Highest Semester SGPA in the Physics Department (Batch of 2023)
  - Semester III, AY 2024–25: SGPA 9.43.
  - Semester IV, AY 2024–25: SGPA 9.09.
  - Semester VI, AY 2025–26: SGPA 9.55.

Internships

1. **IASc-INSA-NASI Summer Research Fellowship Programme 2026**
   Project Supervisor: Prof. Gobinda Majumder and Dr. S Pethuraj
   Institute: Tata Institute of Fundamental Research (TIFR), Mumbai
   Project Title: Correlated and Uncorrelated Noise in Silicon Photomultiplier Devices (SiPM)
2. **IASc-INSA-NASI Summer Research Fellowship Programme 2025**
   Project Supervisor: Dr. Moon Moon Devi
   Institute: Tezpur University, Tezpur, Assam
   Project Title: Particle Physics: Simulations and Machine Learning

Education

2023 - present	Integrated M.Sc at **National Institute of Science Education & Research (NISER)** (Currently in 7th semester)	(CGPA: **9.06**/10)
2023	12th CBSE Board at **Chavara Vidyapeeth, Narsinghpur**	(94.8%)
2021	10th CBSE Board at **Chavara Vidyapeeth, Narsinghpur**	(93%)

Online Courses & Workshops

- Introductory School on the Standard Model of Particle Physics, held virtually in January 2026.
- Quantum Quest: Principles, Security and Applications, organised by AACST.
- Deep Learning, NPTEL course offered by IIT Kharagpur.
- Machine Learning Specialization by Andrew Ng on Coursera.
  - Supervised Machine Learning
  - Advanced Learning Algorithms

Projects

Correlated & Uncorrelated Noise in Silicon Photomultiplier device (SiPM) See details

We experimentally investigate noise contributions in SiPM. Fast output signals are used to study the contributions of pm-OCT, de-CT and after-pulses with variation of SiPM sizes and operating voltage.

Muon Energy Reconstruction in INO-ICAL Detector using Machine Learning See details

We benchmark for 3 ML models: Artificial Neural Network, Random Forest and XGBoost model for energy reconstruction of Muons from a GEANT4 simulated dataset based on INO-ICAL Detector.

Physics Enhanced Neural Networks for Chaotic Hamiltonian Dynamics See details

We implement a Neural network that incorporates the Hamiltonian of the model in its architecture and achieve better prediction stability in phase-space trajectories compared to standard alternatives.

Setting up $\alpha - \gamma$ Coincidence experiment See details

We isolate the decay scheme of ^{241}Am using energy correlation heatmap for coincident events and use time-walk corrected time delay distributions to estimate the lifetime of ^{237}Np excited state.

Research Interests

- Experimental High-Energy Physics
- Scientific Machine Learning
- Detector Development
- Computational Physics

Technical Skills & Relevant Coursework

Programming Languages	Python, C++ (Basic)
Machine Learning & Data Analysis	Experienced in developing Machine Learning models and skilled in data preprocessing, visualization, and performance evaluation.
Software and Tools	LaTeX, Git, Linux, Jupyter Notebook, ROOT
Math Background	Linear Algebra, Probability Theory, Metric Spaces, Differential Equations
Lab experience	Modern Physics, Nuclear Physics and Instrumentation, Solid State Physics and Computational Physics Lab

The data analysis and Lab reports are documented in Github

Extra-Curricular Activities

Sports Cricket & Chess
Club Member of Physics club (Vikiran), NISER

Certificates & Grade Cards

All the certificates and Academic transcript can be accessed via the link below:

Certificates & Grade Cards